ITM University Gwalior

School of Computing and Intelligent Systems

Project-Based Learning (PBL)

Mini Project Report — Submission Format

Course: Data Analytics Using Python

Course Code: CSL0312

Program: B.Tech. CSE (Artificial Intelligence & Machine Learning)

Semester: III **Section:** E

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Project Details

Project Title: Ayushman Bharat Healthcare Analytics: Analysis of Hospital Accessibility and Services

Domain (as per Section 6): Healthcare Analytics

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Date of Submission: 26/10/2026

1. Introduction & Problem Statement

Describe the background, the real-world problem being addressed, its significance, and the scope of the project (200–300 words).

Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (PM-JAY), implemented by the National Health Authority, provides health insurance cover to eligible families at empanelled public and private hospitals. However, holding an Ayushman card does not automatically make care easy to find. Beneficiaries often struggle to know which empanelled hospitals are available in their city or district, what type of hospital each one is, and which specialties or services it offers. Hospital information is usually published as long, unstructured lists, which makes comparison across locations difficult.

This project applies Python-based data analytics to the publicly available Ayushman Bharat hospital empanelment data for Madhya Pradesh. The data is organized into a structured dataset, cleaned, and explored by location, hospital type and specialty information. Tables, summary statistics and charts are used to compare healthcare availability across districts and to identify patterns or areas with comparatively lower coverage.

The problem is significant because clear information on hospital availability helps beneficiaries and their families make better-informed choices, and it helps healthcare planners understand how access varies across locations. The scope of the project is limited to Madhya Pradesh and to the information available in the dataset; it does not measure bed capacity, quality of care or population

served. The results can later be extended into a Smart Health Navigator with filters for State, District, Hospital and Specialty.

2. Dataset Description & Source

Dataset Name: Ayushman Bharat (PM-JAY) Empanelled Hospitals – Madhya Pradesh (Data_sets_of_data_analytics.xls)

Source / Organization: National Health Authority (NHA), PM-JAY Hospital Search Portal

Dataset Link (URL): https://hospitals.pmjay.gov.in/Search/empnlWorkFlow.htm

Type of Dataset: ☒ Public ☐ Private ☐ Self-Collected

Number of Records: 1,230 hospitals

Number of Attributes/Columns: 12 after cleaning (the original file has 18 columns)

Confirm the dataset meets the minimum quality criteria specified in Section 7 of the PBL guidelines: at least 500 records, at least 5 attributes/columns, and minimal missing values (or a documented justification if extensive cleaning was required).

☒ Dataset meets minimum quality criteria (Section 7)

Justification: The dataset has 1,230 records (minimum 500) and 12 attributes (minimum 5). Missing values are minimal: only 22 records (1.8%) have no Hospital Type. Three columns were completely empty and several others held a single value or were not needed, so they were dropped (see Section 3).

Brief description of attributes/columns:

Attribute	Description
Hospital Id	Unique ID of the hospital (used as the key)
Hospital Name	Name of the empanelled hospital
State	State (all records are Madhya Pradesh)
District	District where the hospital is located (51 districts)
Hospital Type	Public, Private (For Profit), Private (Not For Profit), GOI, or Not Specified
Empanelment Type	Scheme under which the hospital is empanelled: PMJAY, Only CGHS, PMJAY and CGHS, PMJAY and CAPF
Convergence Scheme	Whether the hospital is onboarded for the convergence scheme (Yes/No)
Specialities	Specialty codes selected by the hospital (codes starting with M or S, as given in the file; meanings in Section 4)
Num Specialities (derived)	Number of distinct specialty codes listed for the hospital
M Codes (derived)	Number of distinct specialty codes starting with M
S Codes (derived)	Number of distinct specialty codes starting with S
Coverage Level (derived)	High, Medium or Low, based on how many hospitals the district has compared with other districts

3. Data Cleaning & Preprocessing

Document the steps taken to clean and prepare the dataset — handling missing values, duplicates, outliers, data type conversions, encoding, normalization/scaling, etc. Include relevant code snippets or summaries.

The Excel file was loaded into a Pandas DataFrame (1,230 rows, 18 columns) and prepared in the following steps.

1. **Inspection:** checked shape, data types and missing values using df.shape, df.info() and df.isnull().sum().
2. **Dropping unusable columns:** three columns (Unnamed: 5, 6 and 7) were completely empty. Submitted Date and Status Updated Date held only "-NA-", and Application Status had a single value for every hospital, so these carry no information. Upgraded Specialities was missing for 924 records (75%).
3. **Dropping unneeded columns:** Sno (serial number), Hospital Contact (phone numbers, not needed and personal in nature) and Specialities Selected (the Current Specialities column is complete and is used instead).
4. **Text standardization:** removed extra spaces and applied consistent capitalization to Hospital Name and District, so the same district is not counted under different spellings.
5. **Duplicates:** Hospital Id is unique, so no duplicate records were found. 80 rows share the same hospital name and district with another row but have different Hospital Ids; these were kept as separate empanelments (for example branches).
6. **Missing values:** 22 records had no Hospital Type and were labelled "Not Specified" instead of being dropped.
7. **Derived attributes:** Num Specialities, M Codes and S Codes (counts of distinct specialty codes) and Coverage Level (High/Medium/Low from hospitals per district, using the 25th and 75th percentiles of district counts).
8. **Outliers and scaling:** the data is mostly categorical, so encoding and normalization were not required. Very large district counts (Bhopal, 256) were checked and kept because they are genuine values.

Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Dataset file (for .xls files install xlrd once: pip install xlrd)
df = pd.read_excel("Data_sets_of_data_analytics.xls")
print(df.shape)
df.info()
print(df.isnull().sum())
```

```
# Drop empty columns and columns not needed for the analysis
empty = [c for c in df.columns if df[c].isnull().all()]
df = df.drop(columns=empty)
df = df.drop(columns=["Sno", "Hospital Contact", "Specialities Selected",
                      "Upgraded Specialities", "Submitted Date",
                      "Status Updated Date", "Application Status"])
df = df.rename(columns={"Current Specialities": "Specialities",
                        "OnBoarded For Convergence Scheme": "Convergence
Scheme"})
```

```
# Standardize text
for col in ["Hospital Name", "District", "Hospital Type", "Empanelment Type"]:
    df[col] = df[col].astype("string").str.strip()
df["Hospital Name"] = df["Hospital Name"].str.title()
df["District"] = df["District"].str.title()

# Duplicates (Hospital Id is the unique key)
print("Duplicate Hospital Ids:", df.duplicated(subset=["Hospital Id"]).sum())
df = df.drop_duplicates(subset=["Hospital Id"])

# Missing values
print("Missing Hospital Type:", df["Hospital Type"].isnull().sum())
df["Hospital Type"] = df["Hospital Type"].fillna("Not Specified")

# Derived attributes
codes = df["Specialities"].astype(str).str.replace(" ", "").str.split(",")
df["Num Specialities"] = codes.apply(lambda l: len({x for x in l if x}))
df["M Codes"] = codes.apply(lambda l: len({x for x in l if x.startswith("M")}))
df["S Codes"] = codes.apply(lambda l: len({x for x in l if x.startswith("S")}))

dist_counts = df["District"].value_counts()
q1, q3 = dist_counts.quantile(0.25), dist_counts.quantile(0.75)
n = df["District"].map(dist_counts)
df["Coverage Level"] = np.select([n >= q3, n <= q1], ["High", "Low"],
default="Medium")

df.to_csv("ayushman_mp_hospitals_clean.csv", index=False)
print(df.shape)
```

Cleaning summary

Measure	Value
Records before cleaning	1,230
Columns before cleaning	18
Completely empty columns dropped	3
Other columns dropped (constant, sparse or not needed)	7 (Sno, Hospital Contact, Specialities Selected, Upgraded Specialities, Submitted Date, Status Updated Date, Application Status)
Duplicate records (by Hospital Id)	0
Missing Hospital Type (filled as "Not Specified")	22 (1.8%)
Records after cleaning	1,230
Columns after cleaning (including 4 derived)	12

4. Exploratory Data Analysis (EDA)

Summarize the exploratory analysis performed — summary statistics, distributions, correlations, group-wise comparisons, and any patterns identified.

EDA was carried out on the cleaned dataset to understand how hospitals are spread across districts, hospital types, empanelment types and specialties. Because most attributes are categorical, the analysis uses counts, percentages and group-wise comparisons.

```
district_counts = df["District"].value_counts()
print(district_counts.head(10))
print("Districts covered:", df["District"].nunique())

type_counts = df["Hospital Type"].value_counts()
print(type_counts)
print((type_counts / len(df) * 100).round(1))

print(df["Empanelment Type"].value_counts())

print(df["Num Specialities"].describe())
print(df.groupby("Hospital Type")["Num Specialities"].agg(["count", "mean",
"median"]).round(2))

# District vs hospital type (group-wise comparison)
district_type = pd.crosstab(df["District"], df["Hospital Type"])
district_type["Total"] = district_type.sum(axis=1)
top8 = district_type.sort_values("Total", ascending=False).head(8)
print(top8)

# Low-coverage districts
print(district_counts[district_counts <= q1])
```

EDA Measure	Result
Total hospitals analysed	1,230
Districts covered	51
District with the most hospitals	Bhopal (256, 20.8%)
Most common hospital type	Private (For Profit): 611 (49.7%)
Public hospitals	484 (39.3%)
Private (Not For Profit), GOI, Not Specified	95 (7.7%), 18 (1.5%), 22 (1.8%)
Empanelled under PM-JAY (PMJAY, PMJAY and CGHS, PMJAY and CAPF)	1,076 (87.5%)
Only CGHS (not PM-JAY)	154 (12.5%)
Hospitals with 1 or fewer specialty codes	216 (17.6%)

Top 5 districts by number of hospitals

District	Hospitals	Share (%)	Public share (%)
Bhopal	256	20.8	5.5
Indore	114	9.3	17.5
Jabalpur	102	8.3	14.7
Gwalior	80	6.5	10.0

District	Hospitals	Share (%)	Public share (%)
Rewa	28	2.3	60.7

Specialty codes per hospital, by hospital type

Hospital Type	Hospitals	Average codes	Median codes
Private (Not For Profit)	95	8.21	6
Private (For Profit)	611	6.03	4
Public	484	4.95	3
GOI	18	4.83	5
Not Specified	22	1.59	1

Specialty code reference and how common each code is

The Specialities column holds PM-JAY Health Benefit Package (HBP) specialty codes. The meanings below were taken from publicly available state-government HBP lists (see References 6 to 8). Codes starting with M are medical packages and codes starting with S are surgical packages. The counts show how many of the 1,230 hospitals list each code.

Code	Meaning	Hospitals	Share (%)
SB	Orthopaedics	359	29.2
MG	General Medicine	351	28.5
SG	General Surgery (inferred: appendix and gallbladder procedures use SG codes)	309	25.1
SO	Obstetrics & Gynaecology	270	22.0
SU	Urology	268	21.8
MP	Paediatric Medical Management	182	14.8
ER	Emergency-type packages (for example animal bites); full name not confirmed	182	14.8
SN	Neurosurgery	151	12.3
SM	Oral and Maxillofacial Surgery	133	10.8
ST	Polytrauma	131	10.7
SP	Plastic & Reconstructive Surgery	126	10.2
SE	Ophthalmology	117	9.5
SC	Surgical Oncology	108	8.8
MC	Cardiology	107	8.7
SL	Otorhinolaryngology (ENT)	100	8.1
SS	Paediatric Surgery	94	7.6

Code	Meaning	Hospitals	Share (%)
BM	Burns packages	93	7.6
MM	Mental Disorders	6	0.5

Note: The most common codes in the file are M1 (520 hospitals) and M7 (515 hospitals), followed by the numbered series M1 to M8 and S1 to S16. Their meanings could not be confirmed from a reliable public source, so they are reported as codes only. Other letter codes (ID, MN, MO, MR, SV, NA, IN, OT, CP, OC) are also left undecoded. Decoding them is listed under Future Scope.

What this shows: Orthopaedics, general medicine, general surgery, obstetrics and gynaecology, and urology are each offered by more than one in five hospitals, while mental disorders (6 hospitals) and several specialised surgeries are offered by very few.

Patterns identified: Hospitals are concentrated in a few districts: Bhopal, Indore, Jabalpur and Gwalior together hold 552 of the 1,230 hospitals (44.9%), while 14 districts have 9 or fewer. Private for-profit hospitals form the largest group overall, but in smaller districts such as Rewa, Sehore, Raisen and Sagar the public sector is the largest group. Hospitals list 5.7 specialty codes on average (median 4, maximum 30), and not-for-profit private hospitals list the most.

5. Data Visualization & Observations

Insert key charts/plots (bar charts, histograms, scatter plots, heatmaps, etc.) generated using Matplotlib/Seaborn/Plotly, along with a short observation for each visualization.

The charts below were generated with Matplotlib. The code is in the notebook (Section 4) and is shown here.

```
# Figure 1: Top 10 districts
ax = district_counts.head(10).plot(kind="bar", figsize=(9, 5), color="#2F5597")
ax.set_title("Top 10 Districts by Number of Empanelled Hospitals (Madhya
Pradesh)")
ax.set_xlabel("District"); ax.set_ylabel("Number of Hospitals")
plt.xticks(rotation=40, ha="right"); plt.tight_layout()
plt.savefig("fig1_top_districts.png", dpi=150); plt.show()

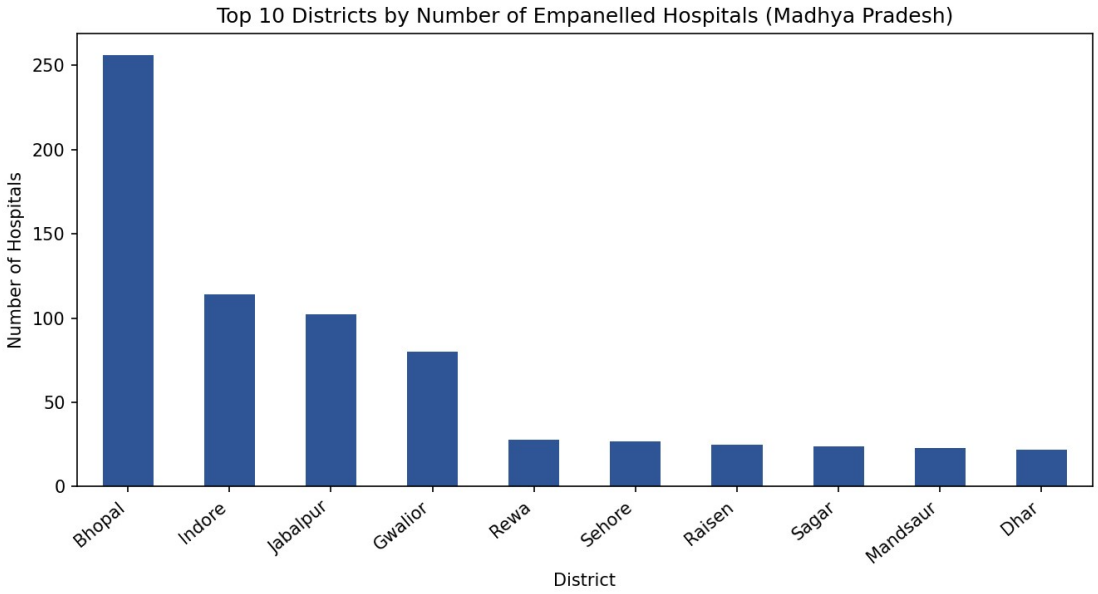
# Figure 2: Hospital type share
type_counts.plot(kind="pie", labels=None, autopct=lambda p: f"{p:.1f}%" if p >= 3
else "",
                 figsize=(6.5, 6), startangle=90, ylabel="")
plt.legend(type_counts.index, loc="lower left", fontsize=8)
plt.title("Hospital Type Distribution"); plt.tight_layout()
plt.savefig("fig2_hospital_type.png", dpi=150); plt.show()

# Figure 3: Distribution of hospitals per district
plt.figure(figsize=(8, 5))
plt.hist(district_counts.values, bins=15, color="#2F5597", edgecolor="white")
plt.title("Distribution of Hospitals per District")
plt.xlabel("Hospitals in a district"); plt.ylabel("Number of districts")
plt.tight_layout(); plt.savefig("fig3_district_hist.png", dpi=150); plt.show()

# Figure 4: Hospital types in the top 8 districts
top8.drop(columns="Total").plot(kind="bar", stacked=True, figsize=(9, 5.5))
plt.title("Hospital Types in the Top 8 Districts")
plt.xlabel("District"); plt.ylabel("Number of Hospitals")
```

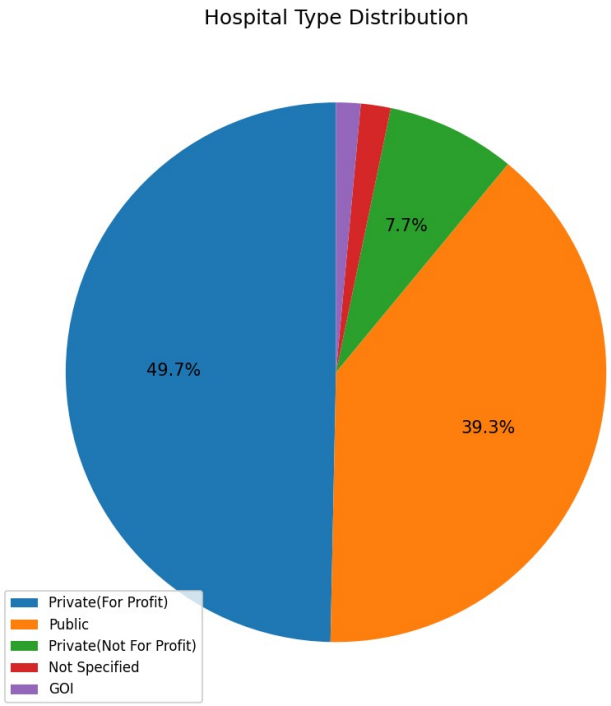
```
plt.xticks(rotation=30, ha="right"); plt.legend(fontsize=8); plt.tight_layout()
plt.savefig("fig4_district_type.png", dpi=150); plt.show()
```

Figure 1: Top 10 districts by number of hospitals



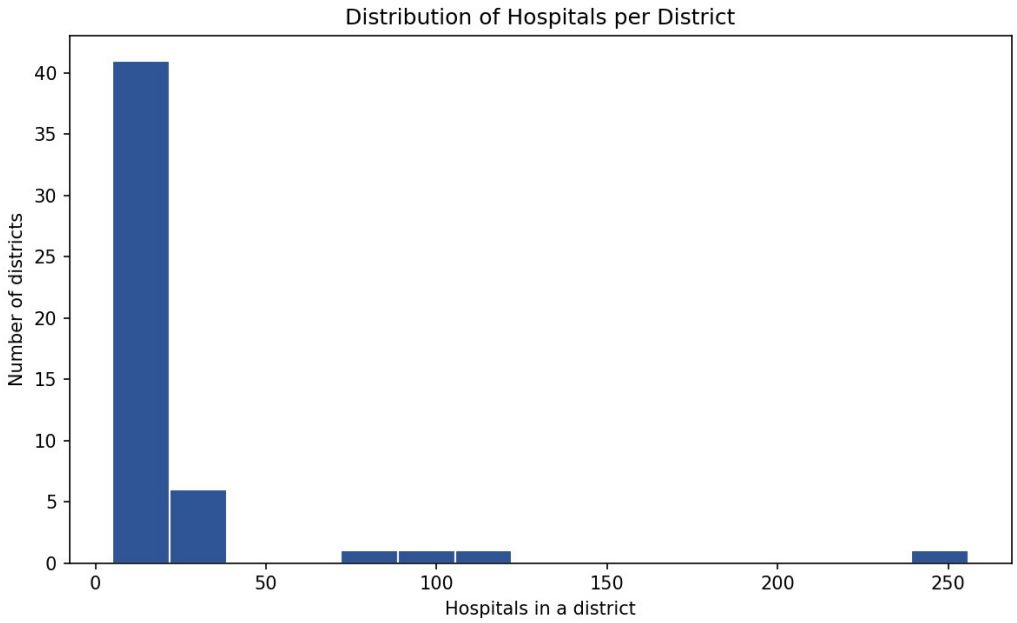
Observation: Bhopal has the most empanelled hospitals (256, or 20.8% of all 1,230), followed by Indore (114), Jabalpur (102) and Gwalior (80). The fifth district, Rewa, has only 28.

Figure 2: Hospital type distribution



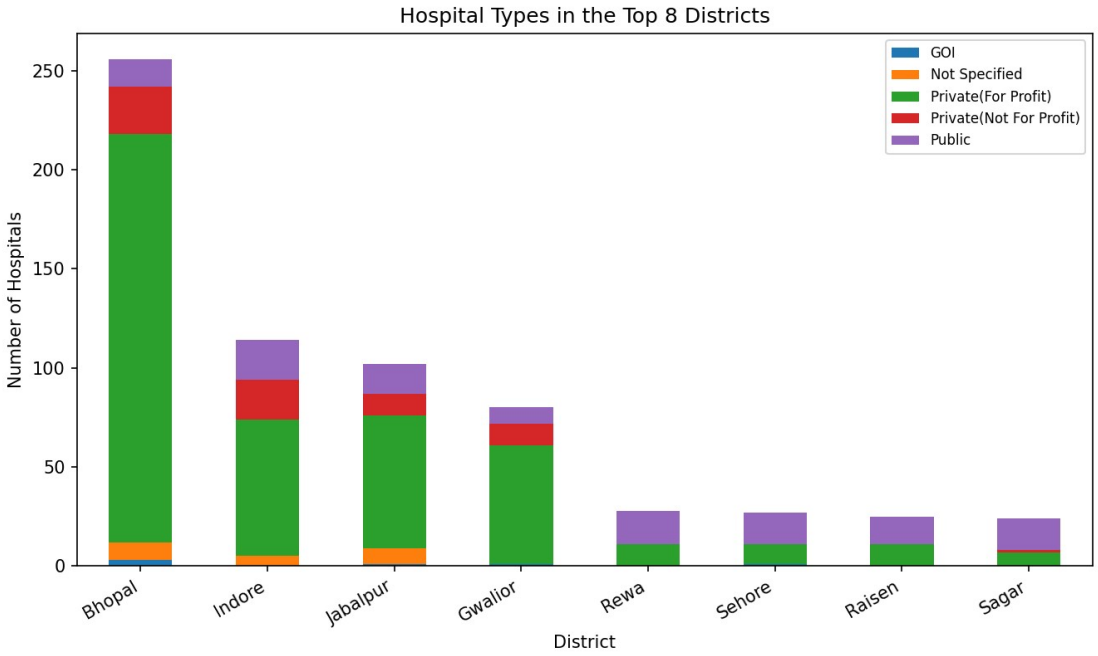
Observation: Private (For Profit) hospitals are the largest group (611, 49.7%), followed by Public (484, 39.3%) and Private (Not For Profit) (95, 7.7%). GOI (18) and unspecified types (22) are small groups.

Figure 3: Distribution of hospitals per district



Observation: The distribution is strongly right-skewed. 47 of the 51 districts have fewer than 30 hospitals (median 14), while four districts are far to the right.

Figure 4: Hospital types in the top 8 districts



Observation: In Bhopal, 206 of 256 hospitals (80.5%) are private for-profit and only 14 (5.5%) are public. In Rewa, Sehore, Raisen and Sagar the public sector is the largest group (for example, 17 of 28 in Rewa).

Figures 5 to 9 add a histogram, a bar plot, a scatter plot, a correlation heatmap and a 100% stacked bar chart.

```
# Figure 5: Histogram of specialty codes per hospital
plt.figure(figsize=(8, 5))
```

```
plt.hist(df["Num Specialities"], bins=range(0, 32, 2), color="#2F5597",
edgecolor="white")
plt.title("Specialty Codes per Hospital")
plt.xlabel("Number of specialty codes"); plt.ylabel("Number of hospitals")
plt.tight_layout(); plt.savefig("fig5_codes_hist.png", dpi=150); plt.show()

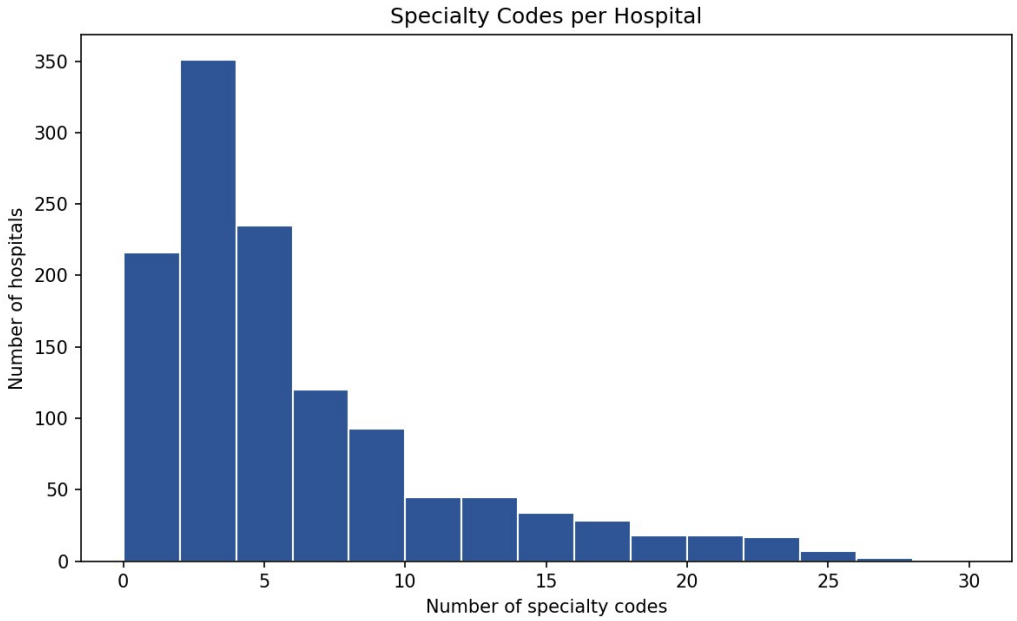
# Figure 6: Bar plot of the most common specialty codes
code_counts = (df["Specialities"].fillna("")
               .apply(lambda s: sorted({c.strip() for c in s.split(",") if
c.strip()}))
               .explode().value_counts())
ax = code_counts.head(10).plot(kind="barh", figsize=(8, 5), color="#2F5597")
ax.invert_yaxis()
plt.title("Top 10 Specialty Codes by Number of Hospitals")
plt.xlabel("Number of hospitals"); plt.ylabel("Specialty code")
plt.tight_layout(); plt.savefig("fig6_top_codes.png", dpi=150); plt.show()

# Figure 7: Scatter plot, hospitals in a district vs average specialty codes
district_stats = df.groupby("District").agg(Hospitals=("Hospital Id", "count"),
                                            AvgCodes=("Num Specialities",
"mean"))
plt.figure(figsize=(8, 5.5))
plt.scatter(district_stats["Hospitals"], district_stats["AvgCodes"],
color="#2F5597")
for name, row in district_stats.sort_values("Hospitals",
ascending=False).head(4).iterrows():
    plt.annotate(name, (row["Hospitals"], row["AvgCodes"]),
                 textcoords="offset points", xytext=(5, 5))
plt.title("Hospitals in a District vs Average Specialty Codes per Hospital")
plt.xlabel("Hospitals in the district"); plt.ylabel("Average specialty codes per
hospital")
plt.tight_layout(); plt.savefig("fig7_scatter.png", dpi=150); plt.show()

# Figure 8: Correlation heatmap
df["District Hospitals"] = df["District"].map(district_counts)
corr = df[["Num Specialities", "M Codes", "S Codes", "District
Hospitals"]].corr()
fig, ax = plt.subplots(figsize=(6.5, 5.5))
im = ax.imshow(corr, cmap="RdBu_r", vmin=-1, vmax=1)
labels = list(corr.columns)
ax.set_xticks(range(len(labels))); ax.set_xticklabels(labels, rotation=30,
ha="right")
ax.set_yticks(range(len(labels))); ax.set_yticklabels(labels)
for i in range(len(labels)):
    for j in range(len(labels)):
        ax.text(j, i, f"{corr.iloc[i, j]:.2f}", ha="center", va="center")
plt.colorbar(im); plt.title("Correlation Heatmap (Pearson)")
plt.tight_layout(); plt.savefig("fig8_correlation.png", dpi=150); plt.show()

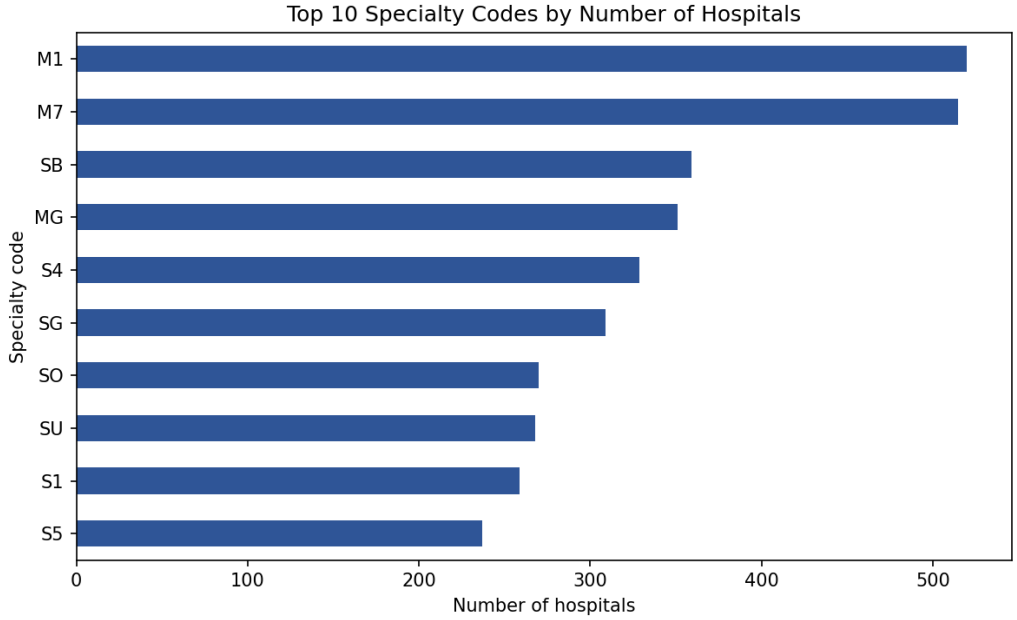
# Figure 9: Hospital type by district coverage level (100% stacked bar)
share = pd.crosstab(df["Coverage Level"], df["Hospital Type"], normalize="index")
* 100
share = share.loc[["Low", "Medium", "High"]]
share.plot(kind="bar", stacked=True, figsize=(8, 5.5))
plt.title("Hospital Type Share by District Coverage Level")
plt.xlabel("District coverage level"); plt.ylabel("Share of hospitals (%)")
plt.xticks(rotation=0); plt.ylim(0, 118); plt.legend(fontsize=8, ncol=3,
loc="upper center")
plt.tight_layout(); plt.savefig("fig9_stacked_share.png", dpi=150); plt.show()
```

Figure 5: Histogram of specialty codes per hospital



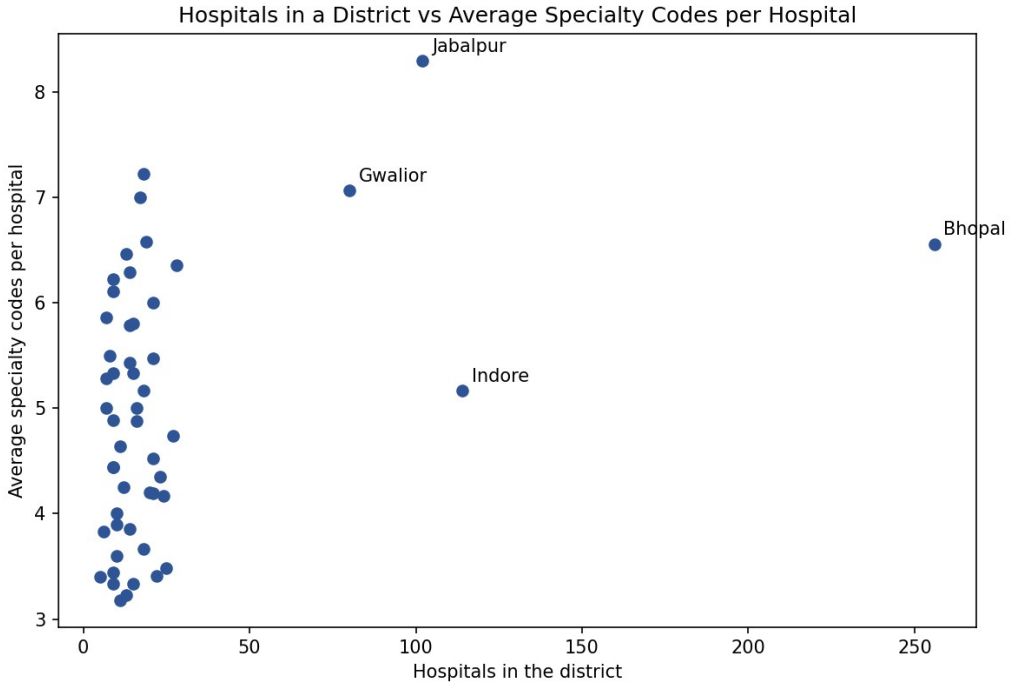
Observation: Most hospitals list few codes: 216 list one code or fewer, 586 list 2 to 5, and 215 list 10 or more. The maximum is 30 codes. The distribution is right-skewed, with a small group of large hospitals listing many specialties.

Figure 6: Top 10 specialty codes by number of hospitals



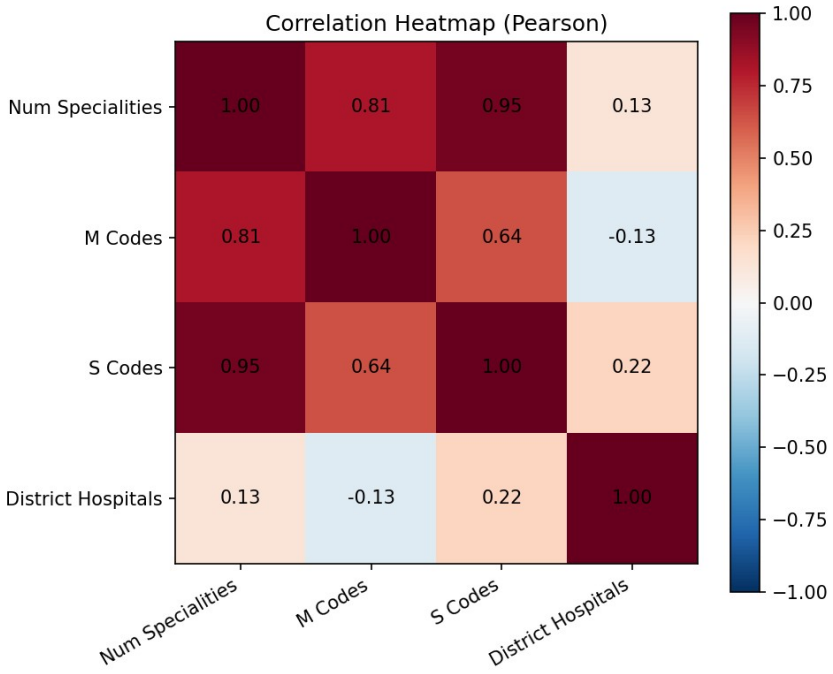
Observation: M1 (520 hospitals) and M7 (515) are the most common codes; their meanings could not be confirmed. Among decoded codes, SB Orthopaedics (359) and MG General Medicine (351) lead, followed by SG, SO and SU.

Figure 7: Scatter plot: hospitals in a district vs average specialty codes per hospital



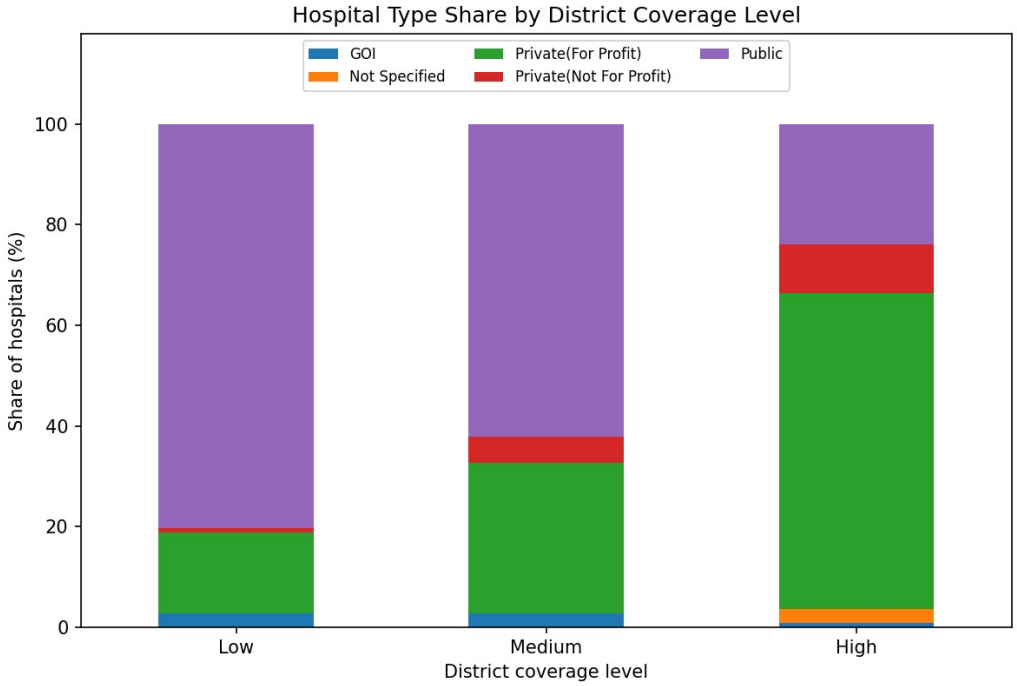
Observation: Across the 51 districts, districts with more hospitals tend to have slightly more specialty codes per hospital (r = 0.357), but the pattern is weak and driven by a few large districts. Jabalpur has the highest average (8.29), Indore (5.17) is below several small districts, and the lowest averages (about 3.2 to 3.4) are in Tikamgarh, Balaghat, Morena, Dindori and Ashok Nagar.

Figure 8: Correlation heatmap (Pearson)



Observation: Num Specialities is strongly correlated with S Codes (0.96) and M Codes (0.81), which is expected because it is built from them. Hospitals in bigger districts show only a weak link with the number of codes (0.13 overall, 0.22 for S codes and -0.13 for M codes).

Figure 9: Hospital type share by district coverage level (100% stacked bar)



Observation: Public hospitals make up 80.4% of hospitals in Low-coverage districts (112 hospitals), 62.2% in Medium (333) and 23.8% in High (785). Private for-profit hospitals rise from about 16% in Low to 30% in Medium and 63% in High-coverage districts.

6. Statistical Analysis (Basic)

Present basic statistical interpretation — measures of central tendency and spread, correlation coefficients, hypothesis tests, or other relevant statistical techniques used to support the analysis.

Since the data is mainly categorical, the statistics focus on the number of hospitals per district (51 districts): its central tendency (mean, median, mode), spread (standard deviation, coefficient of variation) and how unevenly hospitals are distributed (share of the top 3 districts and the Gini coefficient).

```
counts = district_counts.values.astype(float)

print("Mean   :", round(counts.mean(), 2))
print("Median :", np.median(counts))
print("Mode   :", district_counts.mode().iloc[0])
print("Std Dev:", round(counts.std(ddof=1), 2))
print("CV (%) :", round(counts.std(ddof=1) / counts.mean() * 100, 1))
print("Top 3 share (%):", round(counts[:3].sum() / counts.sum() * 100, 1))

# Gini coefficient (0 = equal, 1 = very unequal)
x = np.sort(counts)
k = len(x)
gini = (2 * np.sum(np.arange(1, k + 1) * x) / (k * x.sum())) - (k + 1) / k
print("Gini   :", round(gini, 3))
```

Statistic (hospitals per district)	Value
Mean	24.12

Statistic (hospitals per district)	Value
Median	14
Mode	9
Standard deviation	39.27
Coefficient of variation (%)	162.8
Minimum / Maximum	5 (Ashok Nagar) / 256 (Bhopal)
Share of top 3 districts (%)	38.4
Gini coefficient	0.497

Interpretation: The mean (24.1) is much higher than the median (14), and the coefficient of variation is very large (162.8%), which shows that a few districts pull the average up. The most common district size is just 9 hospitals. The top 3 districts hold 38.4% of all hospitals, and the Gini coefficient of 0.497 indicates a fairly unequal spread of hospitals across districts. These counts are not adjusted for population, so they show availability, not hospitals per person.

Correlation analysis

Pearson correlation coefficients were calculated between the number of specialty codes of a hospital (Num Specialities), its medical (M) and surgical (S) code counts, and the number of hospitals in its district (District Hospitals). A district-level correlation was also calculated between the number of hospitals in a district and the average number of specialty codes per hospital.

```
# Correlation coefficients (Pearson)
print(corr.round(3))

# District level: hospitals in the district vs average specialty codes per
hospital
print(district_stats.corr().round(3))
print("r (district size vs average codes):",
      round(district_stats["Hospitals"].corr(district_stats["AvgCodes"]), 3))
```

Pair of variables	Pearson r	Strength
Num Specialities and S Codes	0.955	Very strong
Num Specialities and M Codes	0.807	Strong
M Codes and S Codes	0.638	Moderate
Num Specialities and District Hospitals	0.129	Very weak
S Codes and District Hospitals	0.218	Weak
M Codes and District Hospitals	-0.126	Very weak (negative)
District size and average codes per hospital (51 districts)	0.357	Weak to moderate

Interpretation: The strong correlations between Num Specialities and the M and S counts are expected, because Num Specialities is calculated from them. The useful result is that a hospital in a

large district does not list many more specialty codes than one in a small district (r about 0.13). At district level the link is weak to moderate (r = 0.357) and depends on a few large districts such as Bhopal, Jabalpur and Gwalior. Correlation does not show cause, and these values describe the listing, not the quality of care.

7. Key Insights & Interpretation

List the key insights derived from the analysis and interpret what they mean in the context of the chosen problem statement.

1. **Hospitals are concentrated in a few districts:** Bhopal alone has 256 hospitals (20.8%), and the top 3 districts hold 38.4% (top 5 hold 47.2%). Beneficiaries in other districts have far fewer empanelled options nearby.
2. **Private sector leads in cities, public sector in smaller districts:** Private (For Profit) hospitals are 49.7% of all hospitals and 80.5% of Bhopal's, while Rewa (60.7%), Sehore, Raisen and Sagar rely mostly on public hospitals. Access outside the big cities depends heavily on public facilities.
3. **Some districts have very low coverage:** 14 districts have 9 or fewer empanelled hospitals, with Ashok Nagar (5), Umaria (6), Alirajpur, Datia and Sheopur (7 each) the lowest. These are candidates for attention from healthcare planners, though population size should be checked before drawing firm conclusions.
4. **Specialty depth is uneven:** hospitals list 5.7 specialty codes on average but the median is 4, and 216 hospitals (17.6%) list one code or fewer. Not-for-profit private hospitals list the most (average 8.2), public hospitals the fewest among the large groups (median 3).
5. **Not every listed hospital is a PM-JAY hospital:** 154 records (12.5%) are empanelled only under CGHS. Beneficiaries need to filter by empanelment type, which supports the idea of a Smart Health Navigator with State, District, Hospital and Specialty filters.

8. Conclusion & Future Scope

Summarize the overall findings of the project and suggest possible extensions or improvements for future work.

Conclusion: This project cleaned and analysed 1,230 empanelled hospital records for Madhya Pradesh across 51 districts. The analysis shows that hospitals are strongly concentrated in a few districts (top 3 hold 38.4%, Gini 0.497), that private for-profit hospitals dominate cities while public hospitals dominate smaller districts, that 14 districts have 9 or fewer hospitals, and that specialty information varies widely between hospitals. These findings give a clear, data-based view of hospital accessibility instead of an unstructured list.

Future Scope:

- Build a Smart Health Navigator (website or app) with filters for State, District, Hospital and Specialty.
- Decode the remaining specialty codes (M1 to M8, S1 to S16 and some letter codes) using the official NHA HBP manual, and analyse availability of individual specialties by district.
- Add population data to compare hospitals per lakh people, and extend the analysis to other states.
- Add map-based views and track changes in empanelment over time.

9. References

List all references, datasets, articles, and tools cited in the project (use a consistent citation style).

1. National Health Authority (NHA), Government of India, PM-JAY Hospital Search Portal (empanelled hospitals list, Madhya Pradesh). https://hospitals.pmjay.gov.in/Search/empnlWorkFlow.htm
2. National Health Authority (NHA), Government of India, "Ayushman Bharat PM-JAY." https://pmjay.gov.in/
3. Government of India, "Open Government Data (OGD) Platform India." https://data.gov.in/
4. HexaHealth, "Ayushman Bharat Hospitals List in Madhya Pradesh" (used for initial topic selection). https://www.hexahealth.com/hospitals/insurance/ayushman-bharat-hospitals-list-madhya-pradesh
5. Pandas and Matplotlib official documentation. https://pandas.pydata.org/docs/ and https://matplotlib.org/stable/
6. Government of Nagaland, CMHIS, list of HBP specialties and codes. https://cmhis.nagaland.gov.in/storage/pageFiles/202.pdf
7. Government of Odisha, GJAY, HBP package listing (General Medicine MG, Paediatric Medical Management MP, Cardiology MC). https://gjay.odisha.gov.in/application/storage/app/uploads/67f9207498dba1744380020.pdf
8. State Health Agency Punjab, order on 25 packages under HBP 2.0 (BM and ER package codes). https://sha.punjab.gov.in/Security_audit/shapunjab/downloadfiles/letters/Order_reagrding_25_packages_HBP20.pdf

10. AI Tool Usage Declaration

As per Section 4 & 15 of the PBL guidelines, any use of AI tools (e.g., ChatGPT, GitHub Copilot, Google Colab AI, Gemini, Claude) must be transparently declared below.

☐ No AI tools were used in this project.

☒ AI tools were used, as detailed below:

AI Tool Used	Purpose (e.g., debugging, documentation)	Extent of Use / Section Applied
ChatGPT (OpenAI)	Drafting the project synopsis and arranging/organizing the data	Synopsis (submitted earlier) and initial data arrangement before analysis
Claude (Anthropic)	Drafting the report text, writing the Pandas/Matplotlib code, running the analysis to produce the numbers and charts, and building the Smart Health Navigator prototype	Report Sections 1–9; notebook code for Sections 3–6; prototype (HTML file)

11. Project Links & Deliverables

GitHub Repository Link: https://github.com/rayeenzaki744-spec/ayushman-bharat-healthcare-analytics

Python Notebook (.ipynb) File Name: ayushman bharat analysis.ipynb

PBL Report File Name: PBL Report Ayushman Bharat Healthcare Analytics.docx

Confirm that the GitHub repository includes the code, notebook, and a PBL report as required.

☒ GitHub repository includes code, notebook, and a PBL report and README

12. Report Structure Checklist (Section 8 Compliance)

S. No.	Report Section (as per Section 8 format)	Included (✓)
1	Title Page	✓
2	Introduction & Problem Statement	✓
3	Dataset Description & Source	✓
4	Data Cleaning & Preprocessing	✓
5	Exploratory Data Analysis (EDA)	✓
6	Data Visualization & Observations	✓
7	Statistical Analysis (Basic)	✓
8	Key Insights & Interpretation	✓
9	Conclusion & Future Scope	✓
10	References	✓

13. Declaration by Student(s)

I/We hereby declare that this Mini Project Report is our own original work, prepared using publicly available/open datasets. All sources of data, references, and AI tool assistance have been duly acknowledged. I/We understand that any instance of plagiarism in code or report will attract strict academic action, as per Section 15 of the PBL guidelines.

Team Leader Signature: _____

Team Member Signature(s): _____

Date: _____

14. For Faculty Use Only — PBL Final Report Evaluation

Evaluation Component B — PBL Final Report Submission | Mid-Term (PBL Based): 10 Marks (Section 10 of PBL Guidelines)

Evaluation Criteria	Max Marks	Marks Awarded
Data Cleaning & Preprocessing	2	
Exploratory Data Analysis (EDA)	2	

Evaluation Criteria	Max Marks	Marks Awarded
Visualization & Observations	2	
Statistical Analysis	2	
Insights, Interpretation & Conclusion	2	
Total (out of 10)	**10**	

Faculty Remarks:

_____

_____

Faculty Mentor Signature: _____

Date: _____